

How SIDEWALL helps protect an F1 car

Think of it as a doctor for four racing tyres. It watches clues, spots trouble, and tells the team when to slow down or pit.

1 • INPUT

Watch the car

Race replay data or a phone driven simulator supplies speed, braking, throttle, position and lap history. The live simulator can also supply wheel and tyre sensors.



2 • SERVER

Clean the clues

FastAPI loads the replay or live session. The feature builder turns the stream into consistent, time ordered measurements.



3 • MONITOR

Look for trouble

Models estimate lock-up, wheelspin, heat and laps left. A virtual tyre sensor estimates heat and pressure where direct measurements are absent. Rules check leaks and flat spots.



4 • DECISION

Choose a call

Health and risk for each tyre become a simple instruction: stay out, manage, box this lap, or box now. The app names the tyre and reason.



5 • PEOPLE

Act in time

The pit wall shows the call and explanation. A WebSocket sends it to the crew and driver phones. The crew can acknowledge it.

Example: Front left becomes too hot → grip may fall → lock-up risk rises → pit wall says MANAGE with driving advice → if danger grows, crew sees BOX THIS LAP.

Replay: browser fetches `/api/replay/{key}` and plays calculated frames. Live: driver phone sends controls over `/ws/driver`; the server streams frames to `/ws/pitwall`; the pit wall relays calls through `/ws/pitwall` to `/ws/crew`. The pit wall sends the call, so its open browser is part of the live alert route.

Evidence boundary: public race telemetry has no tyre pressure, temperature, or wheel speed. Replay values for these are estimates. The debris puncture is a labelled what-if simulation. The system advises a human team; it has not established real-world F1 safety performance.